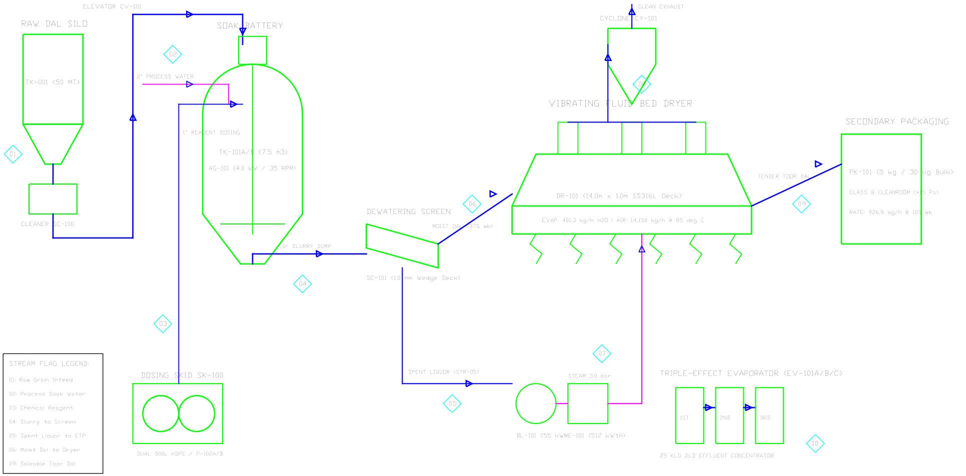
KEY PROCESS PARAMETERS & DESIGN BASIS:

Throughput: 15,000 kg/day as-fed Dal | Operating Schedule: 16.0 h/day (arithmetic mean 937.5 kg/h, design 1,102.9 kg/h @ 85% OEE)

Infeed Moisture: 10.50% w.b. (839.06 kg/h dry solids) | Product Moisture: 10.00% w.b. (926.90 kg/h saleable dal) | Solids Yield: 99.40%

Soak Liquor Ratio: 1.50 : 1.00 (Water:Dal) | Dosing Reagent: Food-Grade NaHCO3 / Citrate | Drying Air: 85 deg C | Deck Area: 14.0 m2

All Product-Contact Metallurgy: AISI SS316L 2B Finish (Ra <= 0.8 um) | CIP / Washdown Compatible | ZLD Effluent Evaporation


ELEVATOR CV-101
RAW DAL SILO
TK-001 (50 MT)
01
CLEANER SC-100
SOAK BATTERY
02
2" PROCESS WATER
1" REAGENT DOSING
TK-101A/B (7.5 m3)
AG-101 (4.0 kW / 35 RPM)
6" SLURRY DUMP
04
03
DEWATERING SCREEN
MOIST DAL (37% wb)
06
SC-101 (1.0 mm Wedge Deck)
CLEAN EXHAUST
CYCLONE CY-101
08
VIBRATING FLUID BED DRYER
DR-101 (14.0m x 1.0m SS316L Deck)
EVAP: 401.3 kg/h H2O | AIR: 14,150 kg/h @ 85 deg C
TENDER TOOR DAL
09
SECONDARY PACKAGING
PK-101 (5 kg / 30 kg Bulk)
CLASS 8 CLEANROOM (+15 Pa)
RATE: 926.9 kg/h @ 10% wb
DOSING SKID SK-100
DUAL 500L HDPE / P-102A/B
SPENT LIQUOR (STR-05)
05
07
STEAM 3.0 bar
BL-101 (55 kWDE-101 (512 kWth)
TRIPLE-EFFECT EVAPORATOR (EV-101A/B/C)
1ST
2ND
3RD
10
25 KLD ZLD EFFLUENT CONCENTRATOR

STREAM FLAG LEGEND:

01: Raw Grain Infeed
02: Process Soak Water
03: Chemical Reagent
04: Slurry to Screen
05: Spent Liquor to ETP
06: Moist Dal to Dryer
09: Saleable Toor Dal

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

PROCESS FLOW DIAGRAM - OVERALL PLANT
DISCIPLINE: PROCESS | STAGE: CONCEPTUAL FEED

DWG NO: DWG-FIN-15T-001
REV: V4.0
SHEET: 1 OF 14
SCALE: AS SHOWN
DATE: 2026-10-15
STATUS: PRELIMINARY
GATE: G-A HOLD